

AN2DL - First Challenge Report

redeagles

Shuvro Ahmed, Umme Humaira Himi, Ahmed Hassan,
Matricola1, 303616 Matricola2, 302484 Matricola3, 305373

November 17, 2025

1 Introduction

Problem definition. In this report the **Pirate Pain Timeseries Classification** challenge: is Addressed. Given a multivariate time-series of body-joint signals recorded per subject, predict their *pain level*—*no_pain*, *low_pain*, or *high_pain*. Each subject also provides *static context* (e.g., number of hands/eyes/legs, possibly prosthetic) and *self-reported pain surveys*. In a formal sense, the following mapping is learned:

$$\begin{aligned} f_{\theta} : (X_{1:T}, s) &\mapsto y \in \mathcal{Y}, \\ X_{1:T} &\in \mathbb{R}^{T \times F}, s \in \mathbb{R}^S, \\ \mathcal{Y} &= \{\text{no_pain}, \text{low_pain}, \text{high_pain}\}. \end{aligned}$$

Goals.

- **Maximise macro-F1** across the three classes.
- **Generalise across subjects** via grouped, stratified cross-validation.
- **Leverage both dynamic (time-series) and static (context/surveys) signals** without data leakage.
- **Deliver a reproducible per-subject inference notebook** suitable for leaderboard submission.csv generation.

Approach overview.

1. *Data splitting.* Applied 5-fold `GroupKFold` using `sample_index` to keep all windows from each sequence in the same fold.
2. *Preprocessing.* Categorical mapping (one/two/three $\rightarrow$ 1/2/3), 1–99 percentile clipping, and a global `StandardScaler` fitted on train only.
3. *Windowing.* Extracted overlapping windows (`SEQ_LEN=100`, `STEP=50`); averaged window-level predictions at inference.
4. *Deep model.* CNN $\rightarrow$ BiGRU $\rightarrow$ Attention encoder with class weighting and early stopping on macro-F1.
5. *Global features.* Computed per-subject mean/std/min/max over all 39 joints and trained a LightGBM classifier.
6. *Ensemble fusion.* Final probabilities obtained via

$$P_{\text{final}} = 0.7 P_{\text{deep}} + 0.3 P_{\text{lgbm}}.$$
7. *Evaluation & test pipeline.* Tracked macro-F1 across folds; reused fold-specific scalers; averaged per-window predictions and exported final CSV.

In summary, the method combines robust preprocessing, window-based deep modeling, global LightGBM statistics, and a tuned ensemble to achieve strong generalization.

2 Problem Analysis

The task is to classify subjects into three pain levels using multivariate time-series motion data plus a few static attributes. Several challenges arise:

- **Variable-length sequences:** recordings differ in duration, requiring fixed-size windowing.
- **Noisy signals:** joint trajectories contain spikes and dropouts that can destabilize training.
- **Class imbalance:** minority pain classes demand macro-F1 rather than accuracy.
- **Mixed modalities:** dynamic joint sequences and static categorical features must be combined effectively.
- **Leakage risk:** overlapping windows from the same subject must not be split across folds.
- **Local vs. global motion:** both fine-grained temporal patterns and overall movement trends influence pain.

3 Method

This section outlines the data preprocessing steps, final architecture and key design choices.

3.1 Preprocessing

- Manual mapping of categorical features (one/two/three $\rightarrow$ 1/2/3), keeping dimensionality low.
- Percentile clipping (1st–99th) on all joint features to remove sensor spikes while preserving global motion trends.
- A global `StandardScaler` was fitted only on the training data and applied to all windows, preventing per-sample leakage.

3.2 Windowing

To convert variable-length sequences into fixed-size tensors, we extracted overlapping windows with:

$$\text{SEQ_LEN} = 100, \quad \text{STEP} = 50.$$

Alternative configurations (50/25, 80/40, 150/75, 200/100) were evaluated but discarded: short windows introduced excessive noise, while long windows produced heavy padding and overfitting. At inference time, window-level probabilities were averaged:

$$P_{\text{sample}} = \frac{1}{K} \sum_{i=1}^K P_{\text{window}_i}.$$

3.3 Neural Architecture: CNN $\rightarrow$ BiGRU $\rightarrow$ Attention

The final sequence model consists of the following layers:

- **Conv1D**(64, kernel size = 5)
- **Conv1D**(128, kernel size = 3)
- **BiGRU**(128, return_sequences = True)
- **Attention mechanism:**
$$\alpha_t = \text{softmax}(w^\top h_t), \quad c = \sum_t \alpha_t h_t$$
- **Dense**(128) + Dropout(0.3) $\rightarrow$ Softmax(3)

CNN layers extract local motion patterns, the BiGRU captures long-range dynamics, and attention highlights the most informative frames[1]. This hybrid model outperformed pure CNN, GRU/LSTM, and Transformer variants. Class weights handled imbalance effectively.

3.4 Global Statistics Model (LightGBM)

For each subject, we computed global statistics over all 39 features (mean, standard deviation, minimum, maximum), resulting in 156 summary features. A LightGBM classifier was trained with: `num_leaves=31`, `max_depth=8`, `n_estimators=500`, `learning_rate=0.05`. Global statistics captured subject-level motion trends that the deep model alone could not, and the two models produced complementary errors.

3.5 Ensemble Fusion

We performed a structured search over blending weights α in:

$$P_{\text{final}} = \alpha P_{\text{deep}} + (1 - \alpha) P_{\text{lgbm}}.$$

Values $\{0.6, 0.65, 0.7, 0.75\}$ were evaluated. A value of $\alpha = 0.7$ provided the best balance between deep and LightGBM predictions, giving the highest macro-F1 and Kaggle performance.

4 Experiments

4.1 Cross-Validation Protocol

We used a 5-fold GroupKFold, grouping all windows by `sample_index` to prevent information leakage. For each fold, we trained the Keras deep model on window-level data, generated out-of-fold (OOF) probabilities, and aggregated them into per-sample predictions by averaging. After training the neural model, LightGBM OOF predictions were computed using a standard KFold, and the two branches were combined using the fusion weight $\alpha = 0.7$. Because each subject produces many overlapping windows, splitting them across train and validation sets would inflate metrics; GroupKFold ensures that all windows from the same sequence stay within the same fold, providing an unbiased estimate of generalization.

5 Results

Comparison between the deep model and the final hybrid ensemble.

Model	Acc	F1	ROC-AUC
CNN-BiGRU-Attention	0.92	0.83	0.95
CNN-BiGRU-Attention+LightGBM	0.93	0.85	0.96

Table 1: Comparison of the two models.

6 Discussion

Strengths. The hybrid CNN-GRU architecture effectively models both local temporal patterns and broader motion structure. The attention mechanism improves interpretability and reduces sensitivity to noisy frames[2]. Ensemble fusion helps mitigate the deep model’s weaknesses on minority classes, and the GroupKFold protocol ensures leakage-free evaluation.

Weaknesses. The dataset is small, and some folds contain difficult edge cases. Global LightGBM features cannot capture fine-grained temporal dynamics, and minority-class OOF performance remains limited.

7 Conclusions

A holistic and reproducible pipeline combining strong preprocessing techniques with windowed dynamic modelling incorporating CNN-BiGRU-Attention architecture is introduced. Global descriptors utilised LightGBM, while a carefully optimised ensemble combination strategy with $\alpha = 0.7$. The hybrid approach performed better than other more complex approaches. The last submission reached a macro F1 value of 0.9427 on Kaggle.

References

- [1] J. Luo, Y. Cao, K. Xie, C. Wen, Y. Ruan, J. Ji, J. He, and W. Zhang. Hybrid cnn-bigru-am model with anomaly detection for nonlinear stock price prediction. *Electronics*, 14(7):1275, 2025.
- [2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.